THE INTEGRAL CLOSURE OF A NORMAL DOMAIN IN A FINITE EXTENSION IS REFLEXIVE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A finite module over a Noetherian normal domain is reflexive if and only if it is the intersection of its localisations at height one primes. Lemma \ref{lemma-reflexive-over-normal}, cited at proof lines 5672-5674. Gives the criterion that the proof verifies.
- The coefficients of the minimal polynomial of an element integral over each height one localisation lie in those localisations. Algebra, Lemma \ref{algebra-lemma-minimal-polynomial-normal-domain}, cited at proof lines 5679-5681. Shows the coefficients lie in every height one localisation.
- A Noetherian normal domain is the intersection of its localisations at height one primes. Algebra, Lemma \ref{algebra-lemma-normal-domain-intersection-localizations-height-1}, cited at proof lines 5681-5683. Brings the coefficients back into the domain.

Source reference. Original lemma 0BM4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14617–14617. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0AVB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13679–13679. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00H7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 633–633. Navigation pointer to a statement the

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Source reference. Original lemma 031T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3996–3996. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let A be a Noetherian normal domain with fraction field K. Let L be a finite extension of K. If the integral closure B of A in L is finite over A, then B is reflexive as an A-module.*

Proof. It suffices to show that $B = \bigcap B_{\mathfrak{p}}$ where the intersection is over height 1 primes $\mathfrak{p} \subset A$, see Lemma 0AVB. Let $b \in \bigcap B_{\mathfrak{p}}$. Let $x^d + a_1 x^{d-1} + \ldots + a_d$ be the minimal polynomial of b over K. We want to show $a_i \in A$. By Algebra, Lemma 00H7 we see that $a_i \in A_{\mathfrak{p}}$ for all i and all height one primes $\mathfrak{p}$. Hence we get what we want from Algebra, Lemma 031T (or the lemma already cited as A is a reflexive module over itself). $\square$